

The Bare Hand

*Pre-PIE *de- as a Free Morpheme, the Directional Particle *de/do, and the Bimorphemic Structure of *deh₃-*

Paper 3 of the der- project. Builds on Chavis & Claude (2026a) and (2026b).

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Chavis, J.M. and Claude, A.I.

Abstract

The first two papers of this project established that pre-PIE *de- underlies the comparative cluster of Paper 1 and the contrastive suffix *-(t)ero- of Paper 2. The present paper goes one step deeper: *der- itself is bimorphemic, decomposing as bare *de- plus locative *-r-. The bare *de- is independently attested as the PIE directional particle *de/do ‘to, toward,’ preserved as the Greek allative clitic -δε (productive in Mycenaean by 1400–1200 BCE), Old Church Slavonic do, Proto-Germanic *tō, and Celtiberian DV at Novallas. The same bare *de- underlies the give-verb *deh₃-, which decomposes as *de- + verbal extension *-h₃-: *der- (the body-part noun) and *deh₃- (the verbal stem) are sibling formations on the same base. The strongest direct evidence for this bimorphemic decomposition is a class-by-class asymmetry in the Greek root-aorist 2pl imperative paradigm: roots the framework analyzes as bimorphemic (including *deh₃- → δότε) preserve a bare-CV form in the 2pl, while roots the framework does not analyze as bimorphemic (including *steh₂- → στητε) preserve the laryngeal-conditioned long vowel throughout. The asymmetry is descriptively established in the standard Greek-grammar tradition; the framework provides a structural account of why it exists. Probability: 35–45%.

1. Introduction

Papers 1 and 2 proposed that a pre-PIE element *de- ‘hand’ underlies both the lexical cluster documented in Paper 1 and the contrastive suffix *-(t)ero- of Paper 2. Each of those arguments required *de- to be reconstructable as a root; neither required it to be attested as an active morpheme in historical IE. The present paper takes the further step of showing that the bare *de- is not merely reconstructable but independently preserved, as the PIE directional particle *de/do ‘to, toward.’

The paper makes two related claims. The first is about **der-*: the body-part noun is bimorphemic, decomposing as bare **de-* plus the locative **-r-* whose status as an independent PIE morpheme is now supported by three published treatments from independent scholarly traditions. The second is about **deh₃-*: the give-verb decomposes as the same bare **de-* plus a verbal extension **-h₃-*, making **der-* (the body-part noun) and **deh₃-* (the verbal stem) sibling formations on the same base rather than the same form. The laryngeal **h₃* is a derivational suffix rather than a root-final consonant, and its addition is post-Anatolian.

The central discriminating evidence for the bimorphemic decomposition of **deh₃-* is a paradigmatic asymmetry in the Greek root-aorist 2pl imperative that the standard handbooks describe but do not explain. The framework predicts the asymmetry structurally; this is the most direct evidence for the bimorphemic claim available anywhere in the comparative record.

The paper proceeds as follows. §2 documents the directional particle **de/do* across the family, with attention to the Mycenaean and Celtiberian attestations. §3 reviews the locative status of **-r-*. §4 develops the bimorphemic decomposition of **deh₃-*. §5 presents the Greek root-aorist 2pl imperative contrast as direct discriminating evidence. §6 addresses the Sanskrit evidence. §7 discusses cross-linguistic typology. §8 sets out scope and remaining problems. §§9–10 close with the remaining paper and the probability assessment.

2. The Directional Particle **de/do* across the Family

PIE **de/do* ‘to, toward’ is a well-attested local particle. Dunkel’s *Lexikon der indogermanischen Partikeln* (2014) treats **de* and **do* as ablaut variants of a single directional particle across the family. Its reflexes include: the Greek allative clitic *-δε*; Old Church Slavonic *do* ‘until, up to, toward’; Proto-Germanic **tō* (English *to*, German *zu*, Dutch *te*); and Celtiberian DV at Novallas. The standard treatment lists **de/do* as a primitive PIE particle with no further decomposition. The framework proposes the missing etymology: **de/do* is the bare descendant of pre-PIE **de-* ‘hand,’ grammaticalized along the pathway HAND > REACH/EXTEND > DIRECTIONAL. The extended hand points; the word for ‘hand’ is reanalyzed as a directional marker. The pathway is typologically well-supported: Heine and Kuteva (2002) document HAND > LOCATIVE/GENITIVE/POSSESSIVE and related developments across Niger-Congo, Austronesian, Sino-Tibetan, and Mayan.

The Greek allative *-δε* and its Mycenaean antiquity. The strongest chronological evidence comes from Greek, where the bare **de-* is preserved as the productive allative clitic *-δε*, marking motion toward a destination: *olkáde* ‘homeward,’ *‘Αθήνῃς* ‘to Athens’ (with sandhi), *pólemonde* ‘to war.’ Linear B tablets from approximately 1400–1200 BCE preserve *-de* as a fully productive

directional clitic on toponyms, with dozens of attested forms (Lejeune 1961; van Hes 2021): *te-qa-de* /T^hēg^wasde/ ‘to Thebes,’ *a-mi-ni-so-de* ‘to Amnisos,’ *da-da-re-jo-de* ‘to the Daidaleion.’ Three features of the Mycenaean attestation are relevant. First, productivity: Mycenaean *-de* is not a closed set of frozen relics but a living clitic applied freely across the corpus. Second, depth of grammaticalization: productive use by 1400 BCE implies that the underlying HAND > DIRECTIONAL development was substantially complete well before the proto-Greek period, placing it plausibly in PIE itself. Third, semantic profile: the clitic is narrowly directional/allative — motion toward, not stative location, not source — exactly the profile expected of a morpheme grammaticalized from ‘where the hand extends.’

The Celtiberian evidence. The Bronze of Novallas (Z.02.01; Beltrán Lloris et al. 2021; Prósper 2022: 19 n.45) attests Celtiberian DV as the directional allative preposition, the Celtiberian reflex of **de/do*. The same inscription preserves **DERNV*[, the Continental Celtic body-part noun documented in Paper 1. That both derivational branches of the bare hand-element — the body-part noun and the directional particle — surface in a single inscription is a convergence not noted in the existing literature, visible only through the framework’s lens.

The Celtiberian institutional vocabulary provides additional evidence in a particularly suggestive register. Prósper (2022: 13) derives Celtiberian *tureita/dureita* as **dū-rextā* ‘issued, directed,’ with the directional prefix **dū-* and the root **reχ-* from PIE **reg-* ‘to direct, rule.’ The preterite *tures* < **dū-reχ-s-t* is independently attested at Torrijo del Campo and at Gortono, and the participial form *DVRETA SALDANICA* appears in peripheral Hispano-Celtic. These forms appear in and on hand-shaped hospitium tesserae cast as clasped right hands (*dextrarum iunctio*). The directional element the framework traces to the pre-PIE hand-root surfaces in Celtiberian both as a standalone allative preposition and as the productive verbal prefix of the institutional vocabulary for right-hand pledge ceremonies — on bronzes cast as clasped hands.

Other branches. OCS *do* ‘until, up to, toward’ is the Slavic reflex. Proto-Germanic **tō* (English *to*, German *zu*, Dutch *te*) is the Germanic reflex, though the Germanic forms admit alternative etymologies and are registered here as supporting rather than load-bearing. The Gaulish compound *de-uor-buet-id* (Lezoux; Lambert 2002, RIG II L-66) independently attests a **de-* preverb in Transalpine Celtic. The SW Palaeohispanic corpus contains **tee-bāre* and related forms that Koch (2019) analyzes as *|de-bāre|* ‘has carried away,’ from Proto-Celtic **dē + √berd-* ‘carry,’ cognate with Latin *dēferō* and Welsh *diferaf*; the Celtic classification of the SW corpus is contested and this evidence is registered conditionally.

A note on Greek *χαμαί*. Greek *χαμαί* ‘on/to the ground’ has figured in debates about whether to reconstruct an allative case for PIE. Melchert (2017) reads it as a fossilized allative **ǵ^hōm-eh₂*

and argues that the Anatolian allative in *-a* descends from PIE **-eh₂*. Beekes (2010) and Frisk (1960–1972) treat the form as a petrified locative. The question remains genuinely open. Paper 4 returns to the Anatolian allative *-a* and its relationship to the directional element in Hittite *dā-*.

3. The Locative **-r-*: Three Independent Witnesses

The locative status of **-r-* is the decompositional argument’s second morpheme. Three independent published treatments from three different scholarly traditions converge on the same conclusion.

Lipp (2009), in reconstructing the PIE heteroclititic *r/n*-stem paradigm, argues that the locative singular forms of acrostatic heteroclitite neuters are built from root nouns combined with an adverbial particle: **-r* and **-én* as parallel locative endings meaning ‘at/in the place of (X).’ On Lipp’s analysis, the spatial semantics is primary and the body-part association is downstream.

Bauhaus (2019) advances a stronger version: **-r* is not an adverbial particle but a true PIE locative case ending — ‘a real case ending, not an Adverbialendung’ — active in pronominal and nominal formations across the family: **k^wor* (Gothic *þor* ‘where’), **tor* (Gothic *þar* ‘there’), nominal locatives such as **ġ^hōm-ér* ‘on the earth.’

Kahl (2024), studying degree morphology in a Harvard dissertation under Jeremy Rau, reaches the same conclusion from a third direction: ‘the origin of the “secondary comparative” of Indo-Iranian and Greek is traced to an archaic locative morpheme **-er*.’ The three treatments are methodologically independent; none is building on the others.

Bauhaus’s most structurally significant contribution for the broader framework is his reconstruction of a paradigmatic opposition between locative **-r* and directive **-o* attached to the same nominal roots: **h₁upér* ‘on top’ vs. **h₁upó* ‘to the top,’ **k(e)nter* vs. **knto*, **ndher* vs. **ndho*. This paradigmatic pair is the structural template the framework reconstructs on **de-*: locative **de-r-* (the body-part noun) versus directive **de + a* (Paper 4’s analysis of Hittite *dā-*). Bauhaus’s observation that the locative **-r* ‘ceased to be productive in all attested languages’ while the directive member ‘could keep its productivity at least until Anatolian’ directly supports the chronological asymmetry the framework reconstructs.

4. **deh₃-* as **de-* + **-h₃-*: the Bimorphemic Decomposition

The framework’s claim about **deh₃-* is morphological rather than phonological. The phonological evidence for **h₃* in the post-Anatolian period is accepted; the question is whether the laryngeal is a root-final consonant of an unanalyzable root, or a derivational extension suffix added to a bare

verbal stem. The framework proposes the latter.

Under this reading, *der- and *deh₃- are sibling formations on the same base. *der- is the nominal derivative: bare *de- plus the locative *-r-, yielding the body-part noun ‘at the place of the hand-direction’ as documented in Papers 1 and 3. *deh₃- is the verbal derivative: bare *de- plus a verbal extension *-h₃-, converting the directional hand-noun into a verbal stem ‘to hand over, to give.’ The extension suffix *-h₃- performs a nominalizing-to-verbal derivational function, parallel to the locative *-r- performing a nominalizing function. The morphological derivation from hand-noun to give-verb via this extension is documented cross-linguistically as one of the most natural grammaticalization pathways from a body-part noun (Heine & Kuteva 2002: 167–168).

The reduplicated present forms — Greek δίδωμι, Sanskrit dádāti — are not a problem for the framework. The i-reduplication belongs to *deh₃-’s inherited morphology and is consistent with a bimorphemic verbal stem; present-stem reduplication operates over the complete stem, whatever its internal structure.

The chronological claim is specific: *-h₃- was added to bare *de- at a post-Anatolian stage. Anatolian split off before the extension was added, which is why Hittite dā- ‘to take’ reflects a pre-extension construction (Paper 4). The post-Anatolian branches inherit *deh₃- already extended. This chronology is independently supported by Norbruis (2021), who argues on semantic grounds that the original meaning of *deh₃- was ‘to take’ and that the ‘give’ meaning is a post-Anatolian innovation.

5. The Greek Root-Aorist 2pl Imperative: Direct Evidence for Bimorphemic *deh₃-

The bimorphemic decomposition of *deh₃- predicts a specific empirical asymmetry in the inherited Greek paradigm. Roots the framework analyzes as bimorphemic — a bare CV-stem plus a laryngeal extension — should preserve the bare pre-extension form in conservative paradigm slots, because that stem was independently anchored in pre-PIE. Roots the framework does not analyze as bimorphemic should show the laryngeal’s full effect throughout, because there is no pre-extension form to anchor against leveling. The Greek root-aorist 2pl imperative is exactly such a conservative slot, and it shows the asymmetry the framework predicts.

PIE root	Framework analysis	2sg aor. impv.	2pl aor. impv.	2pl vowel
*deh ₃ - ‘give’	bimorphemic: *de- + *-h ₃ -	δός	δότε	short o
*d ^h eh ₁ - ‘put’	bimorphemic: *d ^h e- + *-h ₁ -	θές	θέτε	short e

PIE root	Framework analysis	2sg aor. impv.	2pl aor. impv.	2pl vowel
*yeh ₁ -/seh ₁ - ‘let go’	bimorphemic	ἔς	ἔτε	short e
*steh ₂ - ‘stand’	monomorphemic root	στῆθι	στῆτε	long η
*ǵnh ₃ - ‘know’	monomorphemic root	γνῶθι	γνῶτε	long ω
*g ^w eh ₂ - ‘go’	monomorphemic root	βῆθι	βῆτε	long η

Table 1. The Greek root-aorist 2pl imperative contrast. A horizontal rule separates Class A (bimorphemic, short vowel in 2pl) from Class B (monomorphemic, long vowel throughout). The asymmetry is descriptively established in the standard Greek-grammar tradition (Goodell 1902 §368; Smyth §§682–691; Sihler 1995 §502).

The pattern in Table 1 is uniform. The three Class A roots — those the framework analyzes as bimorphemic — preserve a bare-CV form in the 2pl with no laryngeal effect on vowel length. The three Class B roots preserve the laryngeal-conditioned long vowel throughout the paradigm. The standard view’s account is that Class B roots underwent Greek-internal paradigm leveling that extended the long vowel from singular into plural, while Class A roots preserved the original ablaut alternation. This description is accurate as far as it goes, but it leaves a structural question unanswered: why should leveling have operated on one class and not the other? The standard treatments offer no answer.

The framework’s account is structural. Class A roots have bimorphemic structure: their bare-CV pre-extension forms were independently anchored at pre-PIE depth — for *deh₃- specifically, by *de-’s status as a productive directional particle (§2 above), as a body-part noun (Paper 1), and as the source of the contrastive suffix *-(t)ero- (Paper 2). When Greek inherited the paradigm, the bare-CV weak-stem form had multiple structural anchors holding it in place against leveling pressure. Class B roots have no pre-extension form available: *steh₂- ‘stand,’ *ǵnh₃- ‘know,’ *g^weh₂- ‘go’ do not admit bimorphemic decompositions into independently attested bare-CV stems. When leveling pressure operated, the long-vowel singular stem extended into the plural without resistance. The framework predicts the class-by-class distribution; the standard view records it as an unmotivated fact.

The control roots. The cleaner controls for testing the framework’s prediction are *ǵnh₃- ‘know’ (single onset *ǵn-, gives γνῶτε with long ω) and *g^weh₂- ‘go’ (single onset *g^w-, gives βῆτε with long η). Both are single-onset CVH roots that the framework does not analyze as bimorphemic, and both show the laryngeal’s full effect in the 2pl. The contrast therefore tracks the framework’s bimorphemic/monomorphemic distinction rather than any feature of onset shape. *steh₂- ‘stand,’ with its cluster onset *st-, is a less clean control because the long-vowel

preservation could in principle be conditioned by the onset; the two single-onset Class B roots make the point unambiguously.

What the test shows and does not show. The test is not evidence that the standard monomorphemic reconstruction of *deh₃- is incorrect: the standard view's description of the morphological history is accurate and consistent with the framework's account. The framework's advantage is etymological: it explains why Class A and Class B roots should behave differently under leveling pressure, where the standard view leaves the difference as an inherited paradigmatic fact requiring no structural explanation. A competing account that derives the Greek root-aorist 2pl asymmetry from a phonological or analogical mechanism not requiring bimorphemic decomposition would be a genuine challenge to the framework's use of this evidence; no such account has been proposed, but its possibility should be acknowledged.

6. The Sanskrit Evidence

The Greek contrast does not extend straightforwardly to Sanskrit. Sanskrit root-aorist imperative 2pl forms are leveled to long-vowel throughout the paradigm for both classes: √dā → dāta, √dhā → dhātana, √sthā → sthāta. Sanskrit generalized the long-vowel singular stem across the paradigm in both Class A and Class B verbs. This is a real cost: the bimorphemic anchoring that preserved the bare-CV form in Greek was not sufficient to prevent leveling in Sanskrit.

The Sanskrit witness for the bimorphemic framework is accordingly in the present paradigm rather than the aorist. The reduplicated present dadhāti 'puts' gives 2pl imperative datta (with cluster-simplified geminate), parallel to Greek δίδωμι → δίδοτε. The bare-CV weak-stem pattern is recoverable in Sanskrit in conservative present-paradigm slots; the Greek aorist preserves it more clearly.

The principle that emerges from the comparison is weak-stem conservatism in athematic paradigms for bimorphemic roots: in conservative slots where weak-stem ablaut conditions apply, bimorphemic roots preserve the bare pre-extension form because that form was independently anchored, while monomorphemic roots show the laryngeal's full effect because there is no pre-extension form to preserve. Greek is more conservative in the aorist than Sanskrit; both branches preserve the pattern in the present.

7. Cross-Linguistic Typology

The framework's typological predictions for this paper are two: HAND > DIRECTIONAL (underlying the bare *de- of §2) and HAND > GIVE (underlying *deh₃- of §4). Both are well-attested in the cross-linguistic record. HAND > DIRECTIONAL is a natural specialization

of the documented HAND > LOCATIVE/POSSESSIVE pathway (Heine & Kuteva 2002). HAND > GIVE is documented as one of the most natural verbal grammaticalizations from a body-part noun, attested in Niger-Congo, Austronesian, and Sino-Tibetan languages (Heine & Kuteva 2002: 167–168). The framework does not claim that the typological parallels constitute evidence for the reconstruction; they establish that the reconstructed developments are of attested types, which is a necessary though not sufficient condition.

8. Scope and Remaining Problems

The standard etymology of Greek *-δε* does not unanimously trace it to PIE **de/do*; alternative reconstructions exist, and the standard treatment of **de/do* as a primitive particle without etymology is the conservative position. The framework’s proposal is that the particle has a pre-PIE source; this is an additional claim, not a correction of an established derivation.

The property suffix **-ro-* (as in **rudh-ró-* ‘red’) and the instrument suffix **-tro-/trom* contain **-r-* but lack obvious locative semantics. The framework requires an account of the semantic divergence between the locative and the property/instrument uses of **-r-*. Treating them as distinct morphemes with the same phonological shape is one option; treating the property/instrument function as a downstream development from the locative is another. The question is left open here.

The proposed function of **-h₃-* as a verbal extension suffix is reconstructed on the basis of the framework’s architecture rather than independently demonstrated. The §5 evidence supports the bimorphemic decomposition of **deh₃-*, but does not by itself specify the function of the laryngeal extension — only that it has a structural position that the Greek paradigm treats differently from root-final laryngeals. A fuller account of what **-h₃-* does semantically when added to a nominal stem remains to be developed.

The Novallas convergence of **DERNV[* and *DV* in a single inscription is an observation about what the framework makes visible, not an independent argument for the framework. The primary edition (Beltrán Lloris et al. 2021) is the authority for the *DV* reading; Prósper (2022: 19 n.45) provides a parenthetical cross-reference. The Celtiberian institutional vocabulary section rests on Prósper’s analysis and should be evaluated in that context.

9. The Remaining Paper

Paper 4 (‘The Hand That Takes’) carries the framework into Anatolian. The central proposal is that Hittite *dā-* ‘to take’ preserves a pre-Anatolian directive univerbation **de + a*, where the bare hand-element **de-* combined with a postposed directional element **a* ‘toward the speaker’ and

contracted by hiatus to long $*\bar{a}$. The proposal fits a documented productive class in attested Hittite: the Set 1 directional system in which a local element is followed by the directional particle $*-a$ and contracts to a fixed adverb or preverb (anda < $*en + a$; katta < $*\acute{k}mt + a$; parā < $*pro + a$). The proposal resolves the take/give polarity in Anatolian — the apparently anomalous fact that Anatolian’s reflex of $*deh_3-$ means ‘take’ while post-Anatolian reflexes mean ‘give’ — as a structural consequence of the directional $*a$ ‘toward the speaker’ combining with the hand-element to yield ‘extend toward oneself’ = ‘take.’ Independent semantic support is provided by Norbruis (2021), who reconstructs the same chronological ordering on independent grounds.

10. Probability Assessment

Conditional on both Papers 1 and 2 being correct, we assess the probability of the central thesis of this paper — that $*der-$ is bimorphemic, that $*de-$ is independently attested as the PIE directional particle, and that $*deh_3-$ decomposes as $*de- + *-h_3-$ — at approximately 35–45%.

In favour: the Greek root-aorist 2pl contrast is the strongest single piece of discriminating evidence the framework has at any stratum. It is paradigmatic data, not structural inference: a class-by-class asymmetry in the inherited Greek imperative paradigm that aligns with the bimorphemic decomposition and is unmotivated on the standard monomorphemic view. The Mycenaean $-\delta\epsilon$ attestations establish the productivity of the directional clitic by 1400 BCE, placing the HAND > DIRECTIONAL development firmly in PIE. The Celtiberian Novallas convergence and the three independent treatments of the locative $*-r-$ provide additional structural support. Against: the standard etymology of $-\delta\epsilon$ does not unanimously derive it from $*de/do$; the function of $*-h_3-$ as a verbal extension suffix is reconstructed rather than directly demonstrated; and the Sanskrit aorist evidence does not support the framework as clearly as the Greek, requiring the witness to be located instead in the present paradigm.

The marginal probability for the four-paper project as a whole — integrating conditional probabilities across Papers 1–4 — remains in the range 1–4%, with much of the evidence supporting multiple strata simultaneously. We regard the central claim as more likely wrong than right at that level, and the evidential structure as sufficiently dense and the proposal sufficiently testable to warrant publication for scholarly examination.

Authorship and Contributions

Chavis’s contributions. The hypothesis that the directional particle $*de/do$ — specifically as preserved in Greek $-\delta\epsilon$ — descends from a free-standing pre-PIE hand-element originates with

Chavis. The decision to pursue the bimorphemic decomposition of *der- itself is Chavis's. The decision to make the Greek root-aorist 2pl contrast the load-bearing discriminating evidence for bimorphemic *deh₃- is Chavis's, made after identifying the contrast as the strongest single piece of paradigmatic evidence available to the framework. Chavis directed the Mycenaean -δε argument and the Celtiberian institutional vocabulary section, and made the decision to frame the Novallas convergence of *DERNV[and DV as visible only through the framework's lens. All fundamental decisions about scope, framing, and direction rest with Chavis, who supplies all intellectual continuity across sessions through systematic transcript management.

Claude's contributions. Claude drafted the §5 sequence in four subsections and integrated it between §3 (the locative *-r-) and §6 (the Sanskrit evidence), so that the discriminating evidence supports the bimorphemic decomposition of *deh₃- before the Sanskrit complication is addressed. Claude tightened §2 around the three-part argument (productivity, depth of grammaticalization, semantic profile) from prior threads. Claude located and read Lejeune (1961) and van Hes (2021) on the Mycenaean attestations; Koch (2019) on the SW Celtic *tee preverb; Lambert (1995, 2002) on the Gaulish compound verb; Beltrán Lloris et al. (2021) and Prósper (2022) on the Novallas bronze; Goodell (1902), Smyth, and Sihler (1995) on the Greek root-aorist paradigm; Bauhaus (2019) on the locative-directive paradigmatic pairs; and Norbruis (2021) on the semantic chronology. Claude proposed the structuring of the control-root argument and the bimorphemic-anchoring framing. Claude drafted all prose sections from Chavis's structural and substantive direction. The probability estimate in §10 is Claude's own. Claude is an AI system without persistent memory between sessions; the intellectual continuity of the project is supplied by Chavis through systematic transcript management. The collaboration is genuine. The argument was built together. Its strengths and weaknesses belong to both authors, with the responsibility resting with Chavis.

Select References

- Bauhaus, F. 2019. 'The Proto-Indo-European Suffix *-r Revisited.' In Goldstein, R. et al. (eds.), *Sequencæ, Sv̥-Sequencæ and Singlarity in Phonology*, 13–26. Leiden: Brill.
- Beekes, R.S.P. 2010. *Etymological Dictionary of Greek*. 2 vols. Leiden: Brill.
- Beltrán Lloris, F., C. Jordán Cólera, B. Díaz Ariño, and I. Simón Cornago. 2021. 'The Novallas bronze tablet.' *Journal of Roman Archaeology* 34: 713–733.
- Chavis, J.M. and Claude. 2026a. 'The Hand That Gives.' *LingBuzz* (v33).
- Chavis, J.M. and Claude. 2026b. 'A Better Hand.' *LingBuzz* (v10).
- Chavis, J.M. and Claude. 2026d. 'The Hand That Takes.' *LingBuzz* (v5).

- Dunkel, G.E. 2014. *Lexikon der indogermanischen Partikeln und Pronominalstämme*. 2 vols. Heidelberg: Winter.
- Frisk, H. 1960–1972. *Griechisches Etymologisches Wörterbuch*. 3 vols. Heidelberg: Winter.
- Goodell, T.D. 1902. *A School Grammar of Attic Greek*. New York: D. Appleton.
- Heine, B. and T. Kuteva. 2002. *World Lexicon of Grammaticalization*. Cambridge University Press.
- Kahl, L. 2024. *Diachronic Studies in Indo-European Degree Morphology*. Ph.D. dissertation, Harvard University.
- Koch, J.T. 2019. *Common Ground and Progress on the Celtic of the South-western (SW) Inscriptions*. Research Papers 32. Aberystwyth: University of Wales Centre for Advanced Welsh and Celtic Studies.
- Lambert, P.-Y. 1995. *La langue gauloise*. Paris: Errance.
- Lambert, P.-Y. 2002. *Textes gallo-latins sur instrumentum (RIG II, 2)*. Paris: C.N.R.S.
- Lejeune, M. 1961. ‘Les adverbes grecs en -θεν.’ *Bulletin de la Société de Linguistique de Paris* 56: 1–37.
- Lipp, R. 2009. *Die indogermanischen und einzelsprachlichen Palatale im Indoiranischen*. 2 vols. Heidelberg: Winter.
- Melchert, H.C. 2017. ‘An allative case in Proto-Indo-European?’ In Hansen et al. (eds.), *Usque ad Radices*, 527–540. Copenhagen: Museum Tusculanum Press.
- Norbruis, S. 2021. *Indo-European Origins of Anatolian Morphology and Semantics*. LOT Dissertation 588. Amsterdam: LOT.
- Prósper, B.M. 2022. ‘The sibilant sounds of Hispano-Celtic.’ *Journal of Language Relationship* 20/1–2: 1–24.
- Sihler, A.L. 1995. *New Comparative Grammar of Greek and Latin*. Oxford University Press.
- Smyth, H.W. 1920. *A Greek Grammar for Colleges*. New York: American Book Co.
- Stifter, D. 2022. ‘Contributions to Celtiberian Etymology III. The Bronze of Novallas (Z.02.01).’ *Palaeohispanica* 22: 131–136.
- van Hes, S. 2021. *The Ancient Greek Local Suffixes -δε, -θεν, -θε(ν), -τι, -σε and -τις: Function and Origin*. MA thesis, Leiden University.